

JESTER N. ITLIONG

Computational Physicist • Physics Educator • Scientific Software Developer

Ph.D. Candidate in Physics, Michigan Technological University (defense: July 27, 2026)
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PROFESSIONAL SUMMARY

Ph.D. candidate in Physics with expertise in multiscale molecular simulations of polymers, ionic liquids, and soft materials. Experienced in atomistic and coarse-grained molecular dynamics, statistical mechanics, and high-performance scientific computing to investigate transport, rheological, dielectric, and structural properties of complex polymeric systems. Over 8 years of interdisciplinary computational research experience and over 7 years of university teaching, complemented by open-source scientific software including machine-learning tools and interactive simulations. Research connects molecular interactions, mesoscale organization, and macroscopic material behavior through scalable computational workflows and physics-based modeling.

RESEARCH INTERESTS

- Multiscale modeling of polymers and soft materials
- Coarse-grained molecular dynamics
- Structure–property relationships
- Polymer transport phenomena
- Rheology and relaxation dynamics
- Mesoscale organization in polymeric systems
- Statistical mechanics of soft matter
- Machine learning for materials properties

EDUCATION

Michigan Technological University — Ph.D. in Physics

Jan 2021 – present

Houghton, MI, USA • GPA: 4.00/4.00 • Advisor: Prof. Issei Nakamura

- Dissertation: *Molecular Origins of Viscosity in Ionic and Polymerized Ionic Liquids: Insights from Stockmayer Fluid Simulations* (defense scheduled July 27, 2026)
- Graduate Certificate in Advanced Computational Physics (2023)

De La Salle University — M.S. in Physics

2017

Manila, Philippines • Advisor: Prof. Nelson B. Arboleda, Jr.

- Thesis: atomistic MD of water transport and reverse solute diffusion across polyamide membranes in forward-osmosis dewatering of microalgae

Eulogio “Amang” Rodriguez Institute of Science and Technology — B.S. in Applied Physics (minor: Computer Science)

2014

Manila, Philippines • Cum Laude

RESEARCH EXPERIENCE

Graduate Research Assistant — Michigan Technological University

Jan 2021 – present

Computational soft matter physics • Collaboration with Sandia National Laboratories

- Develop coarse-grained MD models of ionic liquids and polymerized ionic liquids to investigate viscosity, ionic conductivity, dielectric behavior, and chain dynamics
- Perform equilibrium and non-equilibrium LAMMPS simulations; analyze transport with Green–Kubo, Einstein–Helfand, and correlation-function methods
- Design scalable Python/Slurm workflows for automated simulation analysis and large-scale statistical-mechanical calculations on HPC resources
- Validate coarse-grained predictions against experimental viscosity and glass-transition data
- Mentor undergraduate and graduate researchers in molecular simulation, scientific programming, and technical communication

Research Assistant — De La Salle University

2018 – 2020

Manila, Philippines

- Performed atomistic MD of polymer, membrane, and electrolyte systems (GROMACS) to investigate water and solute transport in soft materials
- Developed force-field parameters for polymer systems using density functional theory

TEACHING EXPERIENCE

Graduate Teaching Assistant — Michigan Technological University	2023 – 2024
<i>College Physics I Laboratory; Computational Methods in Physics (~75 students)</i>	
• Facilitated laboratory and computational instruction; trained and supported incoming teaching assistants	
Assistant Professorial Lecturer (Part-time) — De La Salle University	2018 – 2020
<i>Physics, Optics, Electronics, and Modern Physics Laboratories (100+ students/year)</i>	
• Developed active-learning laboratory activities and assessments aligned with course outcomes	
Instructor (Full-time) — EARIST	2017 – 2020
<i>Mechanics; Electromagnetism; Statistical Physics; Quantum Physics; Computational Physics (500+ students)</i>	
• Integrated Python programming and computational methods into advanced physics instruction • Supervised undergraduate research projects from design through presentation	
K–12 Science Teacher and College Instructor — De Ocampo Memorial College	2014 – 2015
<i>Biology, chemistry, and physics; introductory courses for pre-medical undergraduates</i>	

TEACHING EFFECTIVENESS AND MENTORING

Evaluations	Undergraduate Physics Laboratory: 4.72/5.00 (AY 2018–19) and 4.88/5.00 (AY 2019–20), both rated Outstanding
Student feedback	“Teaches with clarity and depth.” • “Turns complicated topics understandable.” • “Very approachable and teaches amazingly.”
Mentoring	Mentored undergraduate and graduate students in molecular simulation, scientific programming, data analysis, technical writing, and presentations; supervised research and capstone projects

SOFTWARE AND OPEN-SOURCE PROJECTS

Ionic-Liquid Property Explorer	2026
<i>il-property-explorer.streamlit.app • github.com/jnitlion/il-property-explorer</i>	
• Machine-learning dashboard predicting viscosity, ionic conductivity, density, and glass-transition temperature of ionic liquids from molecular structure, trained on 65,000+ experimental measurements from the NIST ILThermo database • End-to-end pipeline: automated data harvesting, curation, VFT curve fitting, RDKit per-ion featurization, and quantile gradient-boosting models validated on held-out ion families	
Interactive Stockmayer Fluid Simulator	2026
<i>jnitlion.github.io/projects/stockmayer-simulator.html</i>	
• Browser-based 3D molecular dynamics simulation of dipolar fluids (velocity-Verlet integration, thermostat/barostat, dipole–dipole interactions) for teaching and outreach; plain JavaScript + WebGL	

PUBLICATIONS

- [6] **Itliong, J.N.**, Frischknecht, A.L., Stevens, M.J., and Nakamura, I. (2026). Viscosity, Chain Dynamics, and Ion-pairing in Polymerized Ionic Liquids: Effects of Cation Size and Dipole Mobility. *Submitted to Macromolecules*.
- [5] **Itliong, J.N.**, Frischknecht, A.L., Stevens, M.J., and Nakamura, I. (2025). Stockmayer Fluid Simulations for Viscosity and Glass Transition Temperature of Ionic Liquids. *The Journal of Chemical Physics*, 163(4), 044503. doi:10.1063/5.0268727
- [4] Gao, T., **Itliong, J.**, Kumar, S.P., Hjorth, Z., and Nakamura, I. (2021). Polarization of Ionic Liquid and Polymer and its Implications for Polymerized Ionic Liquids: An Overview Towards a New Theory and Simulation. *Journal of Polymer Science*, 59(21), 2434–2457. doi:10.1002/pol.20210330
- [3] **Itliong, J.N.**, Villagracia, A.R.C., Moreno, J.L.V., et al. (2019). Investigation of reverse ionic diffusion in forward-osmosis-aided dewatering of microalgae: A molecular dynamics study. *Bioresource Technology*, 279, 181–188. doi:10.1016/j.biortech.2019.01.109

- [2] **Itliong, J.N.**, Villagrancia, A.R.C., Ong, H.L., et al. (2019). H₂O Absorptivity on a Fully 4-crosslinked Polyacrylamide Membrane via Density Functional Theory and Monte Carlo Calculations for Draw Solution Recovery in Forward Osmosis. *2019 IEEE 11th HNICEM*, Laoag, Philippines. doi:10.1109/HNICEM48295.2019.9072782
- [1] **Itliong, J.N.**, Villagrancia, A.R.C., Moreno, J.L.V., et al. (2018). Density Functional Theory-based modeling and calculations of a polyamide molecular unit for studying forward-osmosis-dewatering of microalgae. *2018 IEEE 10th HNICEM*, Baguio City, Philippines. doi:10.1109/HNICEM.2018.8666436

SELECTED PRESENTATIONS

- "Role of Cation Size and Dipolar Motion Restriction in the Viscosity of Polymerized Ionic Liquids." *APS Global Physics Summit*, Denver, CO, Mar 2026.
- "Effects of Molecular Size and Polarity on Viscosity Scaling and Ionic Conductivity of Polymerized Ionic Liquids." *APS Global Physics Summit*, San Diego, CA, Mar 2025.
- "Influences of Chain Polarity and Molecular Weight on Ion and Polymer Dynamics in Polymerized Ionic Liquids." *APS March Meeting*, Minneapolis, MN, Mar 2024.
- "Effects of Electric Charge, Dipole Moment, and Molecular Size on the Viscosity of Ionic Liquids." *APS March Meeting*, Las Vegas, NV, Mar 2023.
- "Resource Speaker, APS Conference for Undergraduate Women and Gender Minorities in Physics." *Michigan Technological University*, Jan 2025.

HONORS AND AWARDS

Finalist, Midwest Regional 3-Minute Thesis Competition	2026
<i>Midwestern Association of Graduate Schools, Kansas City, MO</i>	
1st Place, 3-Minute Thesis Competition	2025
<i>Graduate Student Government, Michigan Tech (13 finalists)</i>	
Winner, T-Shirt Design Competition	2025
<i>APS Division of Polymer Physics</i>	
Best Poster Presentation	2023
<i>Graduate Physics Colloquium, Michigan Tech</i>	
Outstanding Faculty Award	2019
<i>College of Arts and Sciences, EARIST (voted by 100+ students)</i>	

LEADERSHIP AND SERVICE

2023 – 2024	Graduate Physics Representative , Graduate Student Government, Michigan Tech — advocated for graduate student welfare; developed a peer-evaluation procedure for the Graduate Physics Colloquium
2025	Volunteer , APS Division of Polymer Physics — membership and conference operations, APS March Meeting 2025
2018 – 2020	Research Coordinator , College of Arts and Sciences, EARIST — organized institution-wide research colloquia; developed a systematic archiving workflow for student research

PROFESSIONAL MEMBERSHIPS

American Physical Society, Division of Polymer Physics (2021 – present) • Women in Physics at Michigan Tech (2021 – present) • Philippine Association of Physics and Science Instructors (2015 – 2020)

TECHNICAL SKILLS

Simulation & modeling	Molecular dynamics, coarse-grained modeling, statistical mechanics, transport phenomena, rheology and viscosity calculations, Monte Carlo methods
Scientific computing	Python, C++, Fortran, MPI/OpenMP, Slurm, Linux, high-performance computing
Simulation software	LAMMPS, GROMACS, OVITO, VMD, Gaussian, Avogadro

Data science & ML	NumPy, SciPy, pandas, scikit-learn, RDKit, matplotlib, Streamlit; gradient-boosting and Gaussian-process models with uncertainty quantification
Teaching technologies	Canvas, Zoom, Microsoft Office, Google Workspace

REFERENCES

Issei Nakamura

Associate Professor of Physics
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Dissertation Advisor

Ranjit Pati

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Collaborator